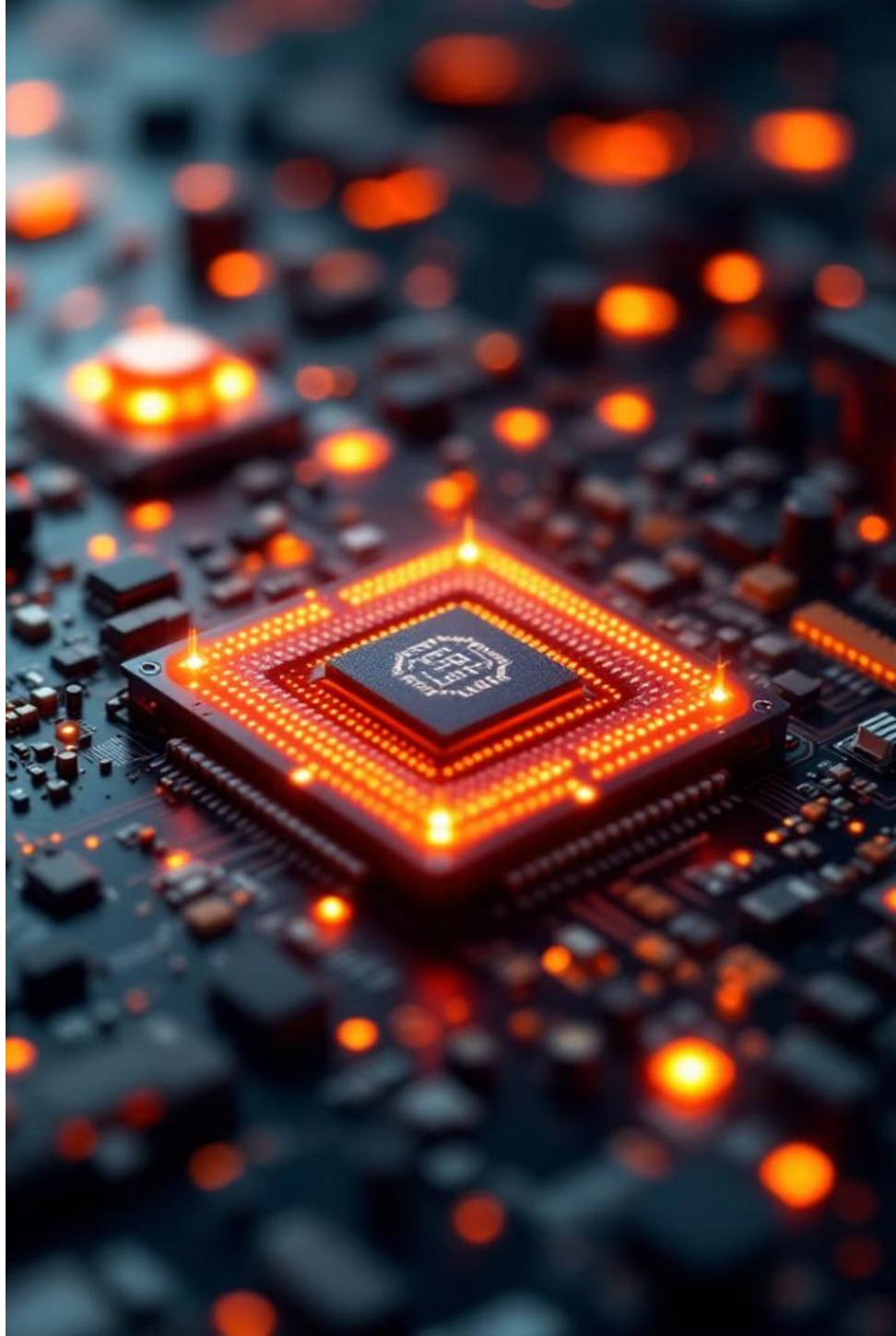
Android Sensors & React Native

Exploring motion, environmental, and position sensors.



Motion Sensors

Accelerometer

Measures acceleration along X, Y, Z axes. Used for step counting and shake detection.

Gyroscope

Measures angular velocity. Used in AR/VR and camera stabilization.

Gravity Sensor

Measures the force of gravity. Used for device orientation.

Linear Acceleration Sensor

Measures acceleration without gravity. Used for motion detection.

Motion Sensors

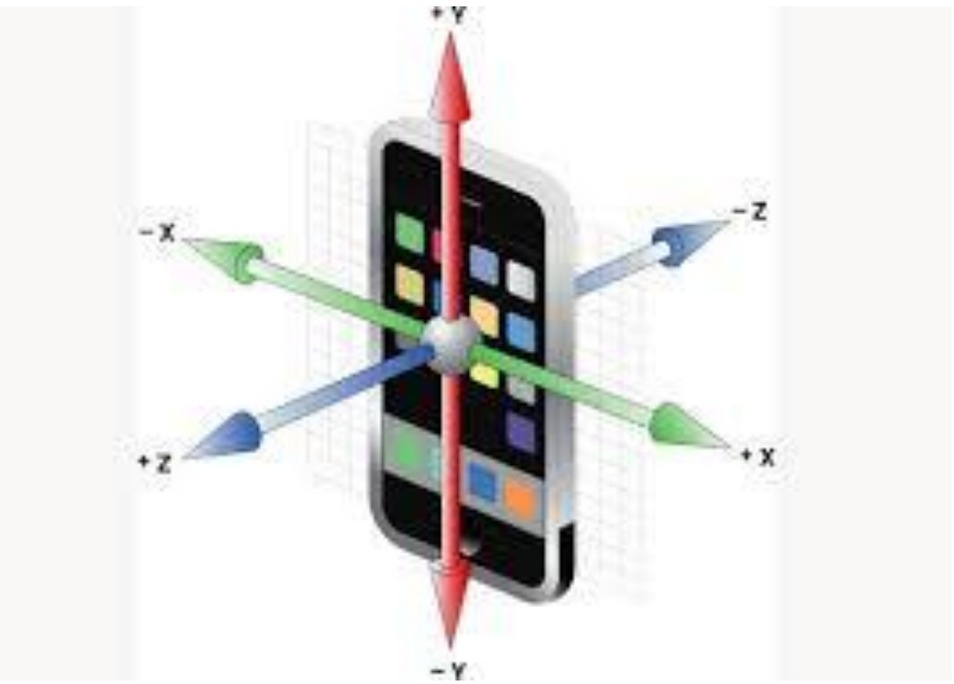
1.1 Accelerometer

- **Function:** Measures acceleration along X, Y, Z axes.
- **Use Cases:**
  - Step counting (e.g., Google Fit, Apple Health)
  - Shake detection for UI interactions (e.g., Snapchat for switching camera)
  - Gaming for tilt-based controls (e.g., Temple Run, Asphalt)
- **Strengths:** Low power consumption, widely available in devices.
- **Weaknesses:** Sensitive to noise, requires filtering for accurate results.

Motion Sensors

1.1 Accelerometer

- Smartphones & Tablets
- Wearable Devices (Smartwatches, Fitness Trackers)
- Laptops
- Automobiles & Motorcycles
- Robots & Drones
- VR & AR Devices
- Medical Applications



Motion Sensors

1.1 Accelerometer

Example:

```
import { accelerometer } from 'react-native-sensors';

accelerometer.subscribe(({ x, y, z }) => {
  console.log(`Acceleration -> X: ${x}, Y: ${y}, Z: ${z}`);
});
```

Motion Sensors

1.2 Gyroscope

- **Function:** Measures angular velocity (rotation) in degrees per second.
- **Use Cases:** AR/VR applications, gaming, motion tracking.
- **Strengths:** Higher accuracy than accelerometer for rotational movement.
- **Weaknesses:** Consumes more power, sensor drift over time.

Motion Sensors

1.2 Gyroscope

- Smartphones & Tablets
- Wearable Devices (Smartwatches, Fitness Trackers)
- Drones & Robotics
- Automobiles
- Cameras & Gimbals
- Aerospace & Aviation
- Medical Applications

Motion Sensors

1.2 Gyroscope

Example:

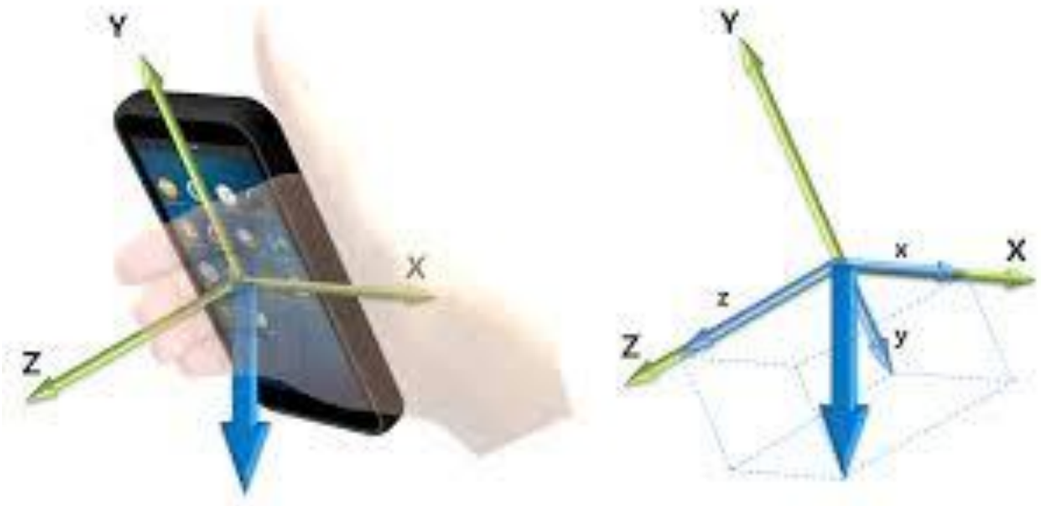
```
import { gyroscope } from 'react-native-sensors';

gyroscope.subscribe(({ x, y, z }) => {
  console.log(`Rotation -> X: ${x}, Y: ${y}, Z: ${z}`);
});
```

Motion Sensors

1.3 Gravity Sensor

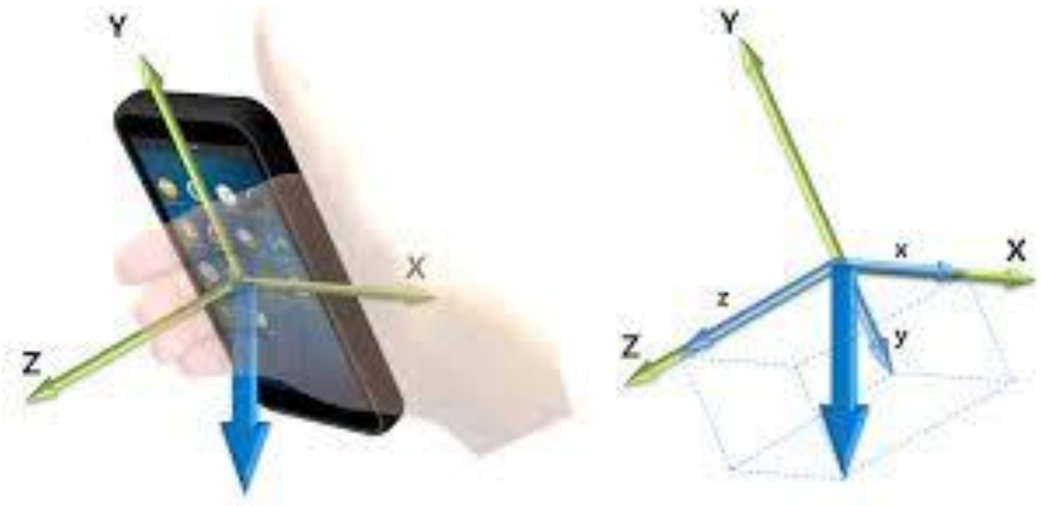
- **Function:** Measures the force of gravity acting on the device.
- **Use Cases:** Device orientation, motion stabilization.
- **Strengths:** More stable than an accelerometer.
- **Weaknesses:** Limited use outside of orientation detection.



Motion Sensors

1.3 Gravity Sensor

- Smartphones & Tablets
- Gaming & VR/AR Applications
- Automobiles & Navigation Systems
- Robotics & Drones
- Medical Applications



Motion Sensors

1.3 Gravity Sensor

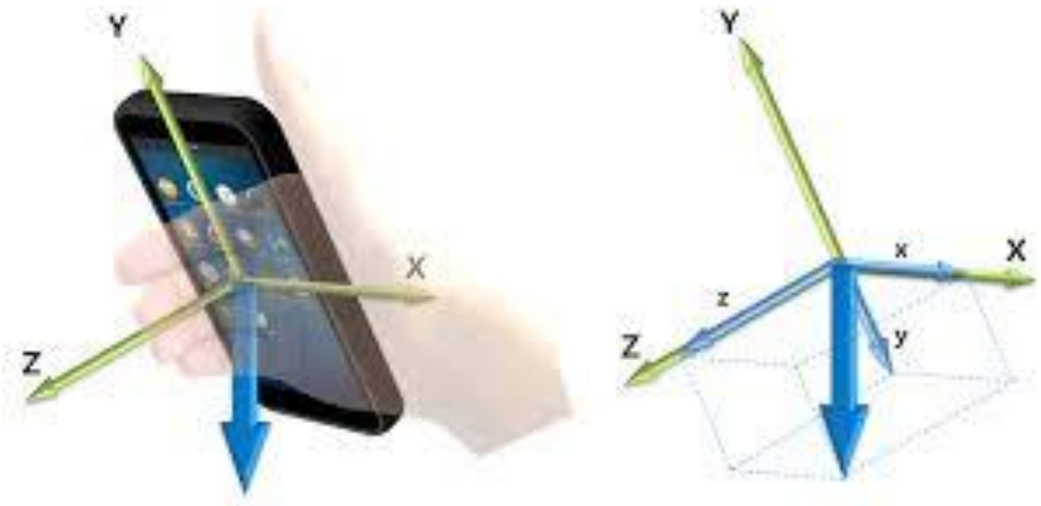
Example:

```
import React, { useState, useEffect } from 'react';
import { View, Text, StyleSheet } from 'react-native';
import { accelerometer } from 'react-native-sensors';

export default function GravitySensorScreen() {
  const [gravity, setGravity] = useState({ x: 0, y: 0, z: 0 });

  useEffect(() => {
    const alpha = 0.8; // Hệ số lọc trọng lực
    let subscription = accelerometer.subscribe(({ x, y, z }) => {
      setGravity((prev) => ({
        x: alpha * prev.x + (1 - alpha) * x,
        y: alpha * prev.y + (1 - alpha) * y,
        z: alpha * prev.z + (1 - alpha) * z,
      }));
    });

    return () => {
      if (subscription) {
        subscription.unsubscribe();
      }
    };
  }, []);
```



Motion Sensors

1.4 Linear Acceleration Sensor

- **Function:** Measures acceleration without gravity interference.
- **Use Cases:** Detecting real-world motion like jogging, jumping.
- **Strengths:** Provides more accurate movement detection.
- **Weaknesses:** Requires post-processing to remove noise.

Motion Sensors

1.4 Linear Acceleration Sensor

- Smartphones & Tablets: Detecting user movement, Enhancing AR & VR experiences
- Navigation & GPS Systems: Supporting GPS when signal is lost, Improving accuracy in digital maps
- Automotive & Transportation Applications: Crash and sudden braking detection, Enhancing cruise control systems, Electronic Stability Program (ESP)
- Robotics & Drones: Flight positioning and stability, Balancing mobile robots
- Sports & Motion Tracking: Sports performance analysis, Training assistance

Motion Sensors

1.4 Linear Acceleration Sensor

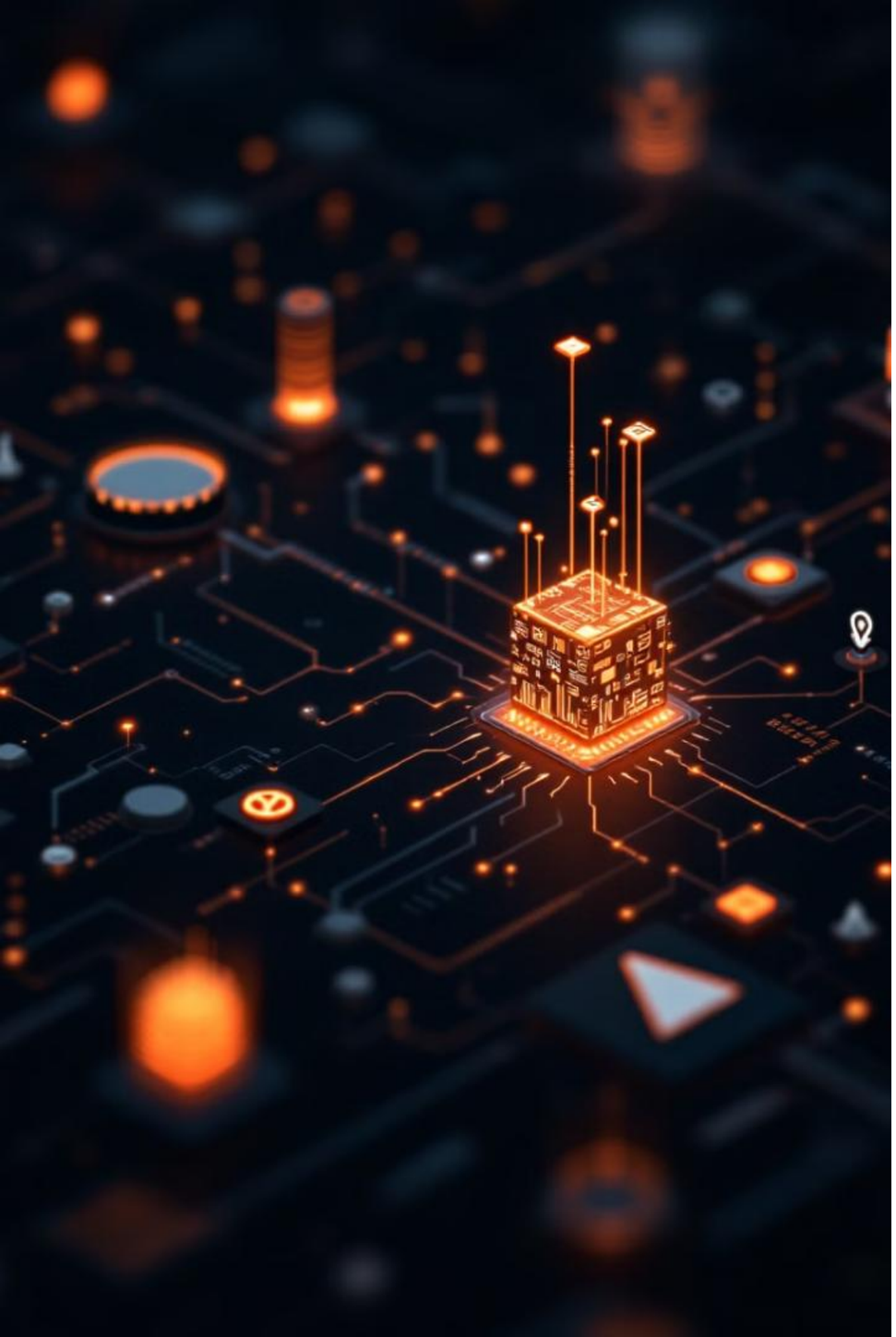
```
import React, { useState, useEffect } from 'react';
import { View, Text, StyleSheet } from 'react-native';
import { accelerometer } from 'react-native-sensors';

export default function LinearAccelerationSensorScreen() {
  const [linearAcceleration, setLinearAcceleration] = useState({ x: 0, y: 0, z: 0 });
  const [gravity, setGravity] = useState({ x: 0, y: 0, z: 0 });

  useEffect(() => {
    const alpha = 0.8; // Filter factor for gravity removal

    const subscription = accelerometer.subscribe(({ x, y, z }) => {
      setGravity((prev) => ({
        x: alpha * prev.x + (1 - alpha) * x,
        y: alpha * prev.y + (1 - alpha) * y,
        z: alpha * prev.z + (1 - alpha) * z,
      }));

      // Subtract gravity to get linear acceleration
      setLinearAcceleration({
        x: x - gravity.x,
        y: y - gravity.y,
        z: z - gravity.z,
      });
    });
```



Environmental Sensors

1 **Light Sensor**

Measures ambient light levels for auto-brightness.

2 **Proximity Sensor**

Detects objects near the screen during calls.

3 **Magnetometer**

Measures magnetic fields for digital compass.

4 **Barometer**

Environmental Sensors

2.1 Light Sensor

- **Function:** Measures ambient light levels in lux.
- **Use Cases:** Auto-brightness adjustment, smart lighting applications.
- **Strengths:** Low power usage.
- **Weaknesses:** Limited precision in certain lighting conditions.

Environmental Sensors

2.1 Light Sensor

- Smartphones & Tablets**:** Automatic screen brightness adjustment, Optimized reading mode
- Laptops & Computer Monitors: Adaptive brightness for energy efficiency, Improved content visibility
- Automobiles & Transportation Systems: Automatic headlight control, Dashboard brightness adjustment
- Smart Home Applications: Automatic lighting control, Smart curtain adjustments
- IoT & Industrial Applications : Workplace lighting regulation, Energy-saving systems in office buildings

Environmental Sensors

2.1 Light Sensor

Example

```
import { light } from 'react-native-sensors';

light.subscribe((illuminance) => {
  console.log(`Light Level: ${illuminance} lux`);
});
```

Environmental Sensors

2.2 Proximity Sensor

- **Function:** Detects objects near the screen.
- **Use Cases:** Auto-turning off the screen during calls, gesture controls.
- **Strengths:** Energy-efficient, useful in phone call scenarios.
- **Weaknesses:** Binary output (near/far), not distance-based.

Environmental Sensors

2.2 Proximity Sensor

- Smartphones & Tablets: Screen turn-off during calls, Pocket mode & accidental touch prevention, Gesture-based controls
- Automobiles & Transportation: Automatic parking assistance, Blind spot detection, Hands-free car unlocking
- Industrial Automation & Robotics: Object detection in assembly, Safety barriers & intrusion detection, Robotics navigation & obstacle avoidance
- Smart Home & IoT Applications: Automatic door opening, Motion-activated lighting, Touchless appliances
- Healthcare & Medical Devices: Contactless patient monitoring, Hand gesture recognition for disabled individuals

Environmental Sensors

2.2 Proximity Sensor

Example

```
import { DeviceEventEmitter } from 'react-native';
import Proximity from 'react-native-proximity';

DeviceEventEmitter.addListener('Proximity', (data) => {
  console.log(data.near ? 'Object detected nearby!' : 'No object detected.');
});
```

Environmental Sensors

2.3 Magnetometer sensor

- **Function:** Measures surrounding magnetic fields.
- **Use Cases:** Digital compass, metal detection, navigation.
- **Strengths:** Works without internet or GPS.
- **Weaknesses:** Affected by electromagnetic interference.

Environmental Sensors

2.3 Magnetometer sensor

- **Navigation & GPS**: Supports digital compasses and improves GPS accuracy.
- **Smartphones**: Enables VR gaming, metal detection, and interference warnings.
- **Automotive**: Assists in car navigation and lane detection.
- **Robotics & Drones**: Helps in navigation, especially in GPS-deprived areas.
- **Security**: Detects hidden electronics and metal weapons.
- **Science**: Measures Earth's magnetic field and aids in mineral exploration.
- **Healthcare**: Powers MRI imaging and motion-tracking devices.

Environmental Sensors

2.3 Magnetometer sensor

Example

```
import { magnetometer } from 'react-native-sensors';

magnetometer.subscribe(({ x, y, z }) => {
  console.log(`Magnetic Field -> X: ${x}, Y: ${y}, Z: ${z}`);
});
```

Environmental Sensors

2.4 Barometer sensor

- **Function:** Measures atmospheric pressure in hPa.
- **Use Cases:** Weather forecasting, altitude measurement.
- **Strengths:** Can be used to improve GPS accuracy.
- **Weaknesses:** Sensitive to temperature changes.

Environmental Sensors

2.4 Barometer sensor

- **Weather Forecasting:** Predicts storms and tracks atmospheric pressure changes.
- **Navigation & GPS:** Improves altitude accuracy for hiking, aviation, and diving.
- **Smartphones & Wearables:** Enhances fitness tracking and location accuracy.
- **Automotive & Aerospace:** Supports tire pressure monitoring and aircraft cabin control.
- **Scientific & Industrial Use:** Helps in climate research and industrial pressure monitoring.
- **Smart Homes & IoT:** Enhances air quality monitoring and weather-adaptive automation.

Environmental Sensors

2.4 Barometer sensor

Example

```
import { barometer } from 'react-native-sensors';

barometer.subscribe((pressure) => {
  console.log(`Pressure: ${pressure} hPa`);
});
```

Position Sensors

GPS

Determines latitude, longitude, and altitude.

Compass

Uses magnetometer to determine direction.

Position Sensors

3.1 GPS (Global Positioning System)

- **Function:** Determines latitude, longitude, and altitude.
- **Use Cases:** Navigation apps, geolocation-based services.
- **Strengths:** High accuracy outdoors.
- **Weaknesses:** High battery consumption, weak performance indoors.

Position Sensors

3.1 GPS (Global Positioning System)

- **Navigation & Mapping**: Helps with driving, cycling, and walking directions.
- **Aviation & Maritime**: Assists pilots and ship captains with navigation.
- **Military & Defense**: Supports missile guidance, reconnaissance, and rescue missions.
- **Transportation & Logistics**: Enhances vehicle tracking and delivery services.
- **Agriculture**: Optimizes farming operations and environmental monitoring.
- **Scientific Research**: Tracks earthquakes and studies wildlife migration.
- **Personal Use**: Helps track devices, fitness activities, and AR gaming.

Position Sensors

3.1 GPS (Global Positioning System)

Example

```
import Geolocation from 'react-native-geolocation-service';

Geolocation.getCurrentPosition(
  (position) => {
    console.log(`Latitude: ${position.coords.latitude}, Longitude: ${position.coords.longitude}`)
  },
  (error) => console.log(error.message),
  { enableHighAccuracy: true, timeout: 15000, maximumAge: 10000 }
);
```

Position Sensors

3.2 Compass

- **Function:** Uses magnetometer to determine direction.
- **Use Cases:** Navigation apps, augmented reality.
- **Strengths:** Works offline.
- **Weaknesses:** Affected by interference from metallic objects.

Position Sensors

3.2 Compass

- **Navigation & GPS**: Enhances location tracking in maps and hiking.
- **Smartphones & AR**: Used in AR maps, gaming, and mobile orientation.
- **Aviation & Maritime**: Helps pilots and sailors with directional guidance.
- **Military & Defense**: Assists with tactical navigation and missile targeting.
- **Automotive & Robotics**: Supports vehicle navigation and autonomous robots.
- **Surveying & Construction**: Helps in land mapping and alignment.
- **Emergency Response**: Aids in search, rescue, and survival situations.

Position Sensors

3.2 Compass

Example

```
import CompassHeading from 'react-native-compass-heading';

CompassHeading.start(10, (degree) => {
  console.log(`Heading: ${degree}°`);
});
```